

Sahil Mohril

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B.Tech Computer Science (Data Science) student with a strong foundation in Java Spring backend architecture, data science methodologies, and interactive visualization. Adept at leveraging core software engineering principles alongside machine learning frameworks to extract actionable insights and solve complex, multi disciplinary problems. Committed to continuous optimization, proactive skill growth, and delivering robust software solutions within collaborative environments.

Education

Vellore Institute of Technology <i>B.Tech in Computer Science and Engineering (Data Science)</i>	CGPA: 9.15	Vellore, India 2023 – 2027
Seth M.R. Jaipuria School Score: 93.5% <i>Higher Secondary School (ISC)</i>		Lucknow, India 2023
Seth M.R. Jaipuria School Score: 96.0% <i>Secondary School (ICSE)</i>		Lucknow, India 2021

Technical Skills

Languages: Java, Python, C, C++, SQL
Web & Backend: Spring Boot, React, HTML, REST APIs
Databases: MySQL, PostgreSQL, MongoDB
Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Exploratory Data Analysis
Big Data & Cloud: Apache Spark (PySpark), Apache Kafka, AWS
Developer Tools: Git, GitHub, Databricks, Tableau, VS Code

Projects

Predictive Maintenance System for Rotating Machinery | *Python, Scikit-learn, Signal Processing* [GitHub]

- Built an end to end predictive maintenance pipeline combining vibration signal processing with machine learning to estimate Remaining Useful Life (RUL) of bearings.
- Engineered time-domain and frequency domain features from raw sensor signals and trained regression models to predict degradation trends.
- Validated pipeline on benchmark bearing run to failure datasets, improving early fault-detection accuracy over baseline threshold methods.

MapReduce Engine in Java | *Java, Multithreading, Distributed Systems* [GitHub]

- Designed and implemented a distributed MapReduce framework from scratch in Java, supporting custom Mapper and Reducer interfaces similar to Hadoop's programming model.
- Built a master-worker architecture to partition input data, schedule map and reduce tasks across worker threads/nodes, and shuffle intermediate key-value pairs.
- Implemented fault tolerance with task re-execution on worker failure and used the engine to run word-count and large-scale aggregation, demonstrating linear speedup with added workers.

CabConnect – Cab Booking Application | *Spring Boot, Java, MySQL* [GitHub]

- Built a full-stack cab booking management platform using Spring Boot, featuring ride booking, user management, and real-time location tracking.
- Designed and exposed RESTful APIs for booking, driver assignment, and trip status updates, integrated with a persistent MySQL backend.
- Implemented role-based access for riders and drivers and structured the backend using layered architecture (Controller-Service-Repository) for maintainability.

Certifications

Oracle Certified Java Professional (SE 11) | *Oracle*

Data Science Course, Cognizance '24 | *IIT Roorkee*

NumPy and Pandas Masterclass | *Udemy*

MySQL Bootcamp | *Udemy*